

SHIFTWM: WORLD MODELS THAT MOVE WHAT THEY SAW

Anonymous authors

Paper under double-blind review

ABSTRACT

World models that forecast in the feature space of a frozen visual foundation model are fast enough for planning, but they must regenerate every future patch from scratch, even though most of a manipulation scene only moves. We introduce SHIFTWM, a latent world model that forecasts by *moving what it has already seen*. For each prefix of future actions, SHIFTWM predicts a local soft transport of the observed patch features, a gate that decides where transport is used, and a learned correction for content that transport cannot explain. Each forecast is computed directly from measured features rather than from the model’s own earlier predictions, so errors do not compound through recursion, and the transport field is a readable estimate of predicted motion. We evaluate on held-out real robot video (DROID), on surgical robot video with recorded kinematics (OpenH-Embodiment, seven dVRK tasks), on single-image manipulation forecasting (IWS), and on standard planning benchmarks (PushT, TwoRoom, Reacher). All comparisons use matched backbones and frozen encoders (DINOv2, DINOv3). We also measure action sensitivity, a property that anchored predictors can lose, and show how an action-contrastive objective recovers it for planning.

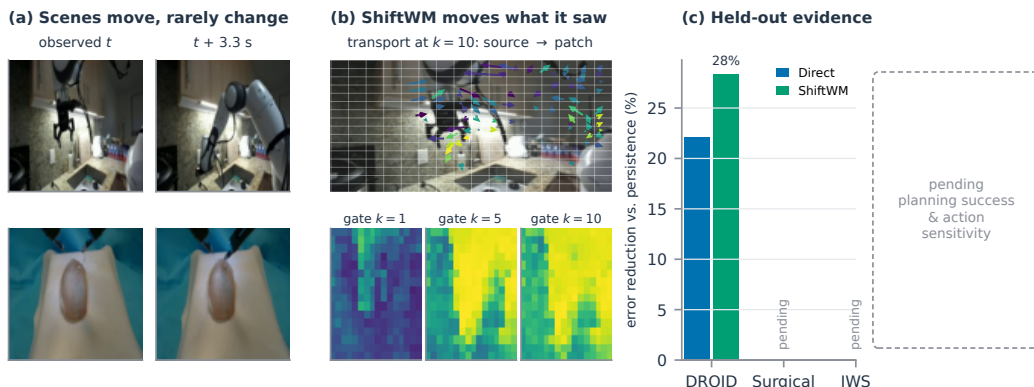


Figure 1: **SHIFTWM forecasts by moving what it has already seen.** (a) In real robot video (DROID, top) and surgical robot video (OpenH, bottom), the scene 3.3 s later mostly contains the same content in new places. (b) SHIFTWM predicts each future patch by transporting observed features along a learned field. Arrows run from the expected source location to the predicted patch on a held-out DROID episode, and the gate maps show that transport is used more at longer horizons and around the moving arm. (c) Held-out gains over persistence across robot, surgical and single-image benchmarks, together with planning success and action sensitivity (Sections 4 and 5).

1 INTRODUCTION

A world model predicts how observations change in response to actions. Recent work shows that forecasting in the feature space of a frozen, self-supervised encoder such as DINOv2 or V-JEPA 2 gives compact models that support planning without generating pixels (Zhou et al., 2025; Assran et al., 2025; Maes et al., 2026b; Sobal et al., 2025). These predictors are *generative in feature space*.

At every future step they output the complete grid of patch features, usually by feeding their own predictions back as inputs. In manipulation and surgical video, however, most of the scene persists: tissue, tools, the table and the robot move, but they are rarely created or destroyed. Regenerating persistent content at every step wastes capacity and lets errors compound under recursion.

We ask a simple question. *Can a latent world model forecast by transporting features it has already observed, and synthesise only what transport cannot explain?* Motion-based video prediction answered a version of this question in pixel space (Finn et al., 2016; Ebert et al., 2017; De Brabandere et al., 2016; Liu et al., 2017). The idea has not been tested for foundation-feature world models used for planning, where features are semantic, high-dimensional and coarse in space. Three questions are open: whether transport helps once the predictor is strong, whether it survives held-out real data, and what it costs for control.

We introduce SHIFTWLM (Figures 1 and 2). A memory encoder reads the patch features of the last few observed frames. For each future step, a decoder conditioned on the causal action-prefix state produces per-patch queries. Each query (i) attends to a local window of *observed* feature locations and copies their content, (ii) gates between staying in place and moving, and (iii) adds a learned correction. Every horizon is predicted directly from measured features, so errors do not compound through recursion, and the transport weights form a motion field that we can inspect and compare against optical flow.

Contributions.

- **Method.** SHIFTWLM, an action-conditioned local feature transport decoder for latent world models. It is a drop-in output head for any frozen patch encoder and adds $< 2\%$ parameters.
- **Evidence across domains.** Matched comparisons against recursive (DINO-WM-style), direct-additive and cross-attention predictors on held-out real robot video (DROID), seven surgical dVRK tasks (Open-H-Embodiment), IWS manipulation forecasting and three planning benchmarks. We use two encoder families and report forecasting, decoded-pixel, probing, action-sensitivity and planning metrics.
- **Analysis.** We show that the learned transport field tracks measured motion. We also characterise a trade-off specific to anchored predictors: open-loop accuracy against action sensitivity. An action-contrastive objective resolves this trade-off for planning.

2 RELATED WORK

World models in foundation-feature and JEPA latent spaces. A recent line of work predicts dynamics in the embedding space of a pretrained or jointly learned encoder instead of in pixels. DINO-WM trains a causal ViT predictor on frozen DINOv2 patch features and plans with model-predictive control (Zhou et al., 2025; Oquab et al., 2024); PLDM, LeWM and JEPA-WMs study joint-embedding predictive models learned from offline trajectories (Sobal et al., 2025; Maes et al., 2026b; Balestriero & LeCun, 2025; Terver et al., 2026), and V-JEPA 2-AC post-trains an action-conditioned predictor on DROID for zero-shot robot planning (Assran et al., 2025; Khazatsky et al., 2024). DINO-world, DINO-Foresight, VFMM and FlowWM forecast future foundation features, deterministically or generatively, and evaluate them through downstream dense heads (Baldassarre et al., 2025; Karypidis et al., 2025; Boduljak et al., 2025; Porcher et al., 2026). DDP-WM separates sparse dynamic regions from a static background, and RLA-WM models residual latent actions over visual features (Yin et al., 2026; Zhang et al., 2026a). Complementary analyses ask which latent spaces and objectives make predictions action-relevant (Nilaksh et al., 2026; Yeom et al., 2026; Zhang et al., 2026b; Huang et al., 2026). In these models the predictor synthesizes each future feature map and, in most cases, conditions later steps on its own outputs. ShiftWM keeps the same frozen-encoder interface, applicable to DINOv2 or DINOv3 features (Siméoni et al., 2025), but changes the output parameterization: every forecast is expressed as a transport of observed features plus a correction, so the predictor is never required to regenerate static content.

Generative world models and model-based reinforcement learning. Pixel-space generative world models learn high-fidelity, action-controllable video simulators. NWM conditions a diffusion transformer on navigation actions (Bar et al., 2025); Cosmos and Cosmos-Predict2.5 provide large

video foundation models for physical AI (NVIDIA, 2025a;b) that Cosmos Policy adapts for control (Kim et al., 2026); Ctrl-World, IRASim, UniSim and IWS target robot manipulation and policy evaluation (Guo et al., 2026b; Zhu et al., 2025; Yang et al., 2024; Wang et al., 2026); and the Genie family generates interactive environments from video, image or text prompts (Bruce et al., 2024; Parker-Holder & Fruchter, 2025). Model-based reinforcement learning instead learns task-specific latent dynamics jointly with rewards and policies, from PlaNet and DreamerV3 to TD-MPC2, IRIS and DIAMOND (Hafner et al., 2019; 2025; Hansen et al., 2024; Micheli et al., 2023; Alonso et al., 2024). These systems optimize sample fidelity or online return with their own encoders, data and, often, billions of parameters, so they do not share our contract of a fixed frozen encoder, identical observations and actions, and targets in common feature coordinates. We therefore treat them as complementary rather than as compute-matched baselines: they address rendering fidelity and online control, whereas we isolate the output parameterization of a compact predictor on a fixed representation.

Transformation- and transport-based prediction. Predicting the future by moving existing content has a long history in video prediction. CDNA and DNA predict action-conditioned convolutional kernels or pixel-motion distributions that transform the previous frame (Finn et al., 2016), dynamic filter networks generate input-dependent filters (De Brabandere et al., 2016), and Vondrick and Torralba transform past pixels with adversarially trained transformers (Vondrick & Torralba, 2017). SNA adds temporal skip connections to the first observed image to handle occlusion (Ebert et al., 2017), while flow-warping approaches synthesize frames by resampling along predicted voxel flow (Liu et al., 2017); SAVP and MCVD extend pixel prediction with stochastic and diffusion models (Lee et al., 2018; Voleti et al., 2022). Recent robot world models reintroduce explicit motion: FlowDreamer predicts 3D scene flow before diffusion rendering (Guo et al., 2026a), AHEAD forecasts VLA patch tokens conditioned on optical-flow velocity (Syed et al., 2026), and anchored navigation world models resist drift through anchor frames and epipolar guidance (Luan et al., 2026). Object-centric models such as SlotFormer and SOLD impose structure through slots (Wu et al., 2023; Mosbach et al., 2025). ShiftWM differs from these designs in four respects. It transports frozen foundation features rather than pixels; its source is a fixed memory of *observed* feature grids that remains unchanged at every horizon, so predicted states are never fed back as inputs; each horizon- h forecast is produced directly from the action prefix $a_{1:h}$; and the transport is a local softmax over a spatial neighborhood, combined with a gated identity path and a learned correction for disocclusion and appearance change.

World models for surgery and evaluation suites. Surgical world modeling is driven by generative video models: Endora simulates endoscopy video (Li et al., 2024), SurgWM learns latent-action-controllable surgical video (Koju et al., 2025), SurgVista targets long-horizon instrument-tissue dynamics (Pan et al., 2026), SWoMo combines neural and symbolic components for cataract surgery (Sivakumar et al., 2026), and Cosmos-H-Surgical learns surgical robot policies from videos through world modeling (He et al., 2025). Open-H-Embodiment releases paired video and kinematics across surgical platforms, together with a kinematics-conditioned surgical simulator (Open-H-Embodiment Consortium, 2026); SutureBot, JIGSAWS, Cholec80 and CholecT50 provide surgical demonstrations and task, skill and workflow annotations (Haworth et al., 2025; Gao et al., 2014; Twinanda et al., 2017; Nwoye et al., 2022), and SurgWMBench benchmarks instrument-motion planning (Liu et al., 2026). We use Open-H dVRK recordings to test kinematics-conditioned forecasting in feature space, a regime these pixel generators do not target. More broadly, robot datasets such as DROID, BridgeData V2 and Open X-Embodiment supply action-labeled video (Khazatsky et al., 2024; Walke et al., 2023; Open X-Embodiment Collaboration, 2024). WorldScore, World-ModelBench, EWMBench and Physics-IQ score generated video for controllability and physical plausibility (Duan et al., 2025; Li et al., 2025; Yue et al., 2025; Motamed et al., 2026), whereas stable-worldmodel standardizes latent planning evaluation (Maes et al., 2026a). We follow the latter protocols for control and report feature-space forecasting error for prediction.

3 METHOD

Problem. A frozen encoder E maps an image to a grid of $N = G \times G$ patch features $\mathbf{Z}_t \in \mathbb{R}^{N \times C}$. Given H observed grids $\mathbf{Z}_{-H+1:0}$, the actions between them $\mathbf{a}_{-H+1:-1}$, and a future action se-

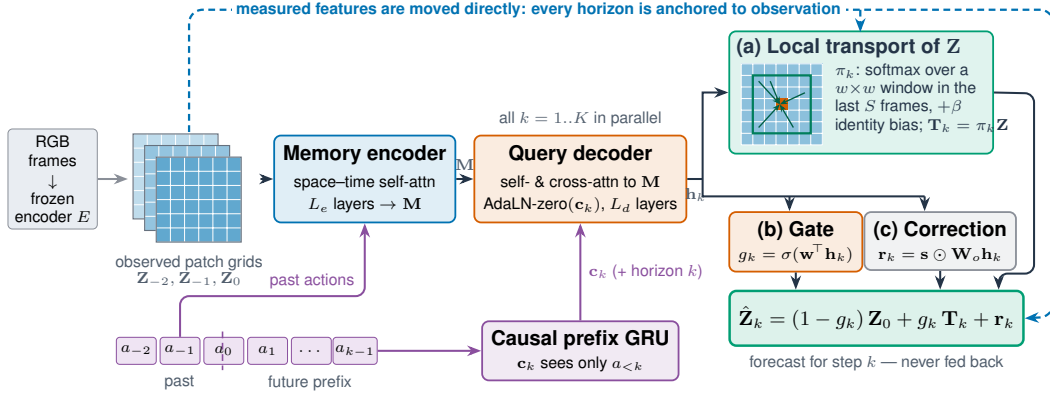


Figure 2: **SHIFTWMM architecture.** A frozen encoder maps the H observed frames to patch grids $\mathbf{Z}_{-H+1:0}$. The memory encoder mixes them over space and time. For each future step k , a decoder conditioned on the causal action-prefix state $\mathbf{c}_k$ produces per-patch hidden states $\mathbf{h}_k$. The output combines three terms: (a) local soft transport of *observed* features, (b) a gate between the identity path and transport, and (c) a scaled correction. Every horizon reads measured features directly, so no forecast is fed back as input.

quence $\mathbf{a}_{0:K-1}$, a world model predicts $\hat{\mathbf{Z}}_{1:K}$. The forecast $\hat{\mathbf{Z}}_k$ may depend only on $\mathbf{a}_{<k}$ (causality). Features are standardised per channel with train-split statistics that are shared across positions.

Memory encoder. Each observed patch feature is projected to width d and receives a spatial embedding, a frame embedding and an embedding of the action applied after that frame. L_e transformer layers over all $H \cdot N$ tokens give memory states $\mathbf{M} \in \mathbb{R}^{H \times N \times d}$.

Action-prefix queries. A unidirectional GRU over embedded actions gives a causal prefix state. We add a horizon embedding to form $\mathbf{c}_k$. Query tokens start from the memory states of the last frame, and L_d decoder layers apply self-attention across patches and cross-attention to $\mathbf{M}$. Each layer is modulated by $\mathbf{c}_k$ through adaptive layer normalisation with zero-initialised gains (Peebles & Xie, 2023). The result is $\mathbf{h}_{k,i} \in \mathbb{R}^d$ for patch i at step k . All K steps are decoded in parallel.

Local feature transport. For query patch i , let $\mathcal{W}(i)$ be the $w \times w$ spatial window around i in each of the last $S \leq H$ observed frames. Transport weights are

$$\pi_{k,i}(j) = \text{softmax}_{j \in \mathcal{W}(i)} \left(\frac{1}{\sqrt{d_k}} (\mathbf{W}_q \mathbf{h}_{k,i})^\top (\mathbf{W}_k \mathbf{M}_j) + \beta \mathbb{I}[j = (0, i)] \right), \quad (1)$$

where β is a learned identity bias toward the same location in the last frame. Transport moves *measured* features, not latent states: $\mathbf{T}_{k,i} = \sum_j \pi_{k,i}(j) \mathbf{Z}_j$.

Gated output. A per-patch gate $g_{k,i} = \sigma(\mathbf{w}_g^\top \mathbf{h}_{k,i} + b_g)$ and a correction $\mathbf{r}_{k,i} = \mathbf{s} \odot \mathbf{W}_o \mathbf{h}_{k,i}$ give

$$\hat{\mathbf{Z}}_{k,i} = (1 - g_{k,i}) \mathbf{Z}_{0,i} + g_{k,i} \mathbf{T}_{k,i} + \mathbf{r}_{k,i}. \quad (2)$$

$\mathbf{W}_o$ is zero-initialised and $b_g = -3$, so an untrained model starts close to persistence and learns departures from it. $\mathbf{s} \in \mathbb{R}^C$ is a learned per-channel scale. Appendix E compares it with the bounded (tanh) correction used in earlier variants.

Training. We minimise the mean squared error over all horizons, $\mathcal{L}_{\text{pred}} = \frac{1}{K} \sum_k \|\hat{\mathbf{Z}}_k - \mathbf{Z}_k\|^2$. Anchoring every forecast to observed features makes it easy to ignore actions: a forecast that stays close to $\mathbf{Z}_0$ scores well on average. For planning we therefore add an action-contrastive hinge. For each sample it compares the prediction error under the true future actions with the error under actions borrowed from another sample in the batch:

$$\mathcal{L}_{\text{act}} = \max(0, m - (\|\hat{\mathbf{Z}}(\tilde{\mathbf{a}}) - \mathbf{Z}\|^2 - \text{sg} \|\hat{\mathbf{Z}}(\mathbf{a}) - \mathbf{Z}\|^2)), \quad \mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{act}}. \quad (3)$$

Matched baselines. To isolate the output head, all learned predictors share the same memory encoder, decoder, width and depth. **Direct** replaces Equation (2) with $\mathbf{Z}_0 + \mathbf{W}_o \mathbf{h}_k$. It has the same cross-patch attention, so it controls for spatial communication. **AR** predicts a residual to the most recent grid and is rolled out recursively, with gradients through the rollout. **AR-TF** follows DINO-WM: it is trained with teacher forcing to predict absolute features and is rolled out at test time. **Persistence** copies $\mathbf{Z}_0$, and **Linear** extrapolates the last feature difference.

Planning. For control we plug each predictor into the same model-predictive planner (CEM) with the official budgets of each benchmark. The cost is the feature distance between the predicted grid and the goal grid (Section 4.4).

4 EXPERIMENTS

We ask four questions. **(Q1)** Does transport improve held-out forecasting over matched predictors on real robot and surgical video (Section 4.2)? **(Q2)** Which components matter, and does the gain hold across encoders and horizons (Section 4.3)? **(Q3)** What happens in closed-loop planning (Section 4.4)? **(Q4)** What does the transport field learn (Section 5)? All numbers come from completed runs with three training seeds. Selection uses validation splits only, and test splits are scored once.

4.1 SETUP

Datasets (Table 6). **DROID** (Khazatsky et al., 2024): 1,126 real Franka episodes from 24 official shards, split by recording session (851/143/132). We train on the primary exterior camera and evaluate transfer to the second exterior camera. One model step is 5 frames (0.33 s) with a 35-D block of Cartesian and gripper commands, and we forecast $K=10$ steps (3.3 s). **Open-H-Embodiment, Hamlyn dVRK** (Open-H-Embodiment Consortium, 2026): 1,013 episodes of seven surgical training tasks recorded with a scene camera. Actions are 80-D blocks of bimanual Cartesian targets, and one step is 1/3 s. Splits are episode-disjoint per task (708/152/153). **IWS** (Wang et al., 2026; Zhang et al., 2026a): single-image forecasting on PushT, Box and Rope, evaluated on the official RLA-WM handles. **Planning:** PushT, TwoRoom and Reacher with the official LeWM/stable-worldmodel protocol (Maes et al., 2026b;a).

Encoders, models and metrics. We use frozen DINOv2-S/14 and DINOv2-B/14 (Oquab et al., 2024) and DINOv3-S/16 (Siméoni et al., 2025), all at 16×16 patches with no pooling. Every learned predictor has width 384 and $L_e=L_d=4$ (≈ 22 M parameters), and all share the optimiser, schedule, step count and batch size (Appendix A). Forecasting metrics are standardised feature MSE (averaged over horizons, and at the final horizon), cosine similarity, and MSE after 4×4 pooling. We also report decoded PSNR, SSIM and LPIPS, linear probes of robot state, and action sensitivity. For planning we report success rate and planning time. Episodes are the statistical unit, and we give paired bootstrap 95% intervals (Appendix K).

4.2 HELD-OUT FORECASTING ON REAL ROBOT AND SURGICAL VIDEO (Q1)

Table 1 and Figure 3 report held-out forecasting. pending

4.3 WHAT MATTERS: COMPONENTS, ENCODERS AND HORIZONS (Q2)

pending

4.4 CLOSED-LOOP PLANNING (Q3)

Using the official protocol, the released LeWM checkpoint reaches 88% on PushT in our harness (mean of 3 planner seeds; the LeWM paper reports 96 ± 2.8 over training seeds). This confirms that our planner is faithful (Appendix H). pending

Table 1: **Held-out forecasting** (standardised feature MSE ↓, DINOv2-S, mean of 3 seeds). Avg” averages horizons 1–10 and K ” is the final step. DROID cam 2 is evaluated zero-shot with models trained on cam 1. Best in **bold**, second underlined. †: paired 95% interval against the best baseline excludes zero. [Interim: seeds completed – DROID: ar 0/3, ar-tf 0/3, direct 1/3, shiftwm 1/3; Hamlyn: ar 0/3, ar-tf 0/3, direct 0/3, shiftwm 0/3.]

Method	DROID cam 1		DROID cam 2 (0-shot)		Open-H Hamlyn		IWS
	Avg	K	Avg	K	Avg	K	
Persistence	0.249	0.326	pending	pending	pending	pending	pending
Linear extrapolation	4.812	11.855	pending	pending	pending	pending	pending
AR-TF (DINO-WM-style)	pending	pending	pending	pending	pending	pending	pending
AR (rollout-trained)	pending	pending	pending	pending	pending	pending	pending
Direct (cross-attn)	<u>0.194</u>	<u>0.248</u>	pending	pending	pending	pending	pending
SHIFTWM (ours)	0.179†	0.229†	pending	pending	pending	pending	pending

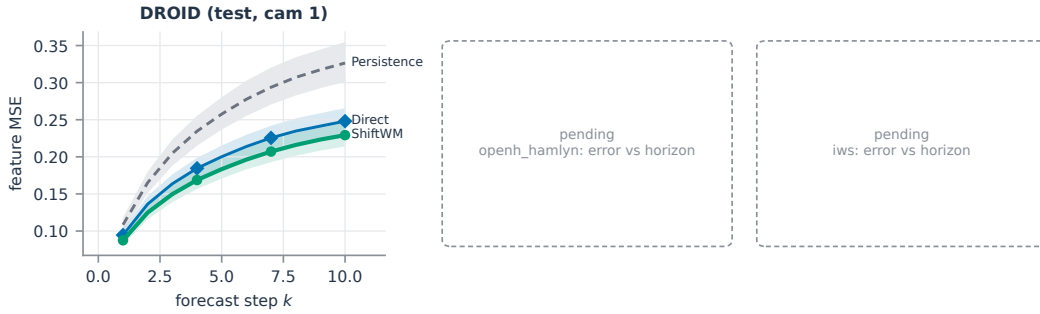


Figure 3: **Error grows more slowly with horizon under transport.** Standardised feature MSE per forecast step on held-out episodes. Recursive predictors (AR, AR-TF) accumulate error. SHIFTWM and Direct read observed features at every step, and SHIFTWM additionally moves them.

5 ANALYSIS

What does the transport field learn? (Q4) We convert $\pi_{k,i}$ into an expected source displacement per patch and compare it, at patch resolution, with optical flow estimated by RAFT (Teed & Deng, 2020) between the observed and future frames (Figure 4). pending

Where do the gains come from? We bin test patches by the magnitude of their true feature change and compare the per-bin error of SHIFTWM and Direct (Appendix I). pending

Accuracy versus action sensitivity. A forecast that stays close to $\mathbf{Z}_0$ can score well on average while barely responding to the commanded actions, and a planner then cannot rank candidate action sequences. We measure this directly as the rate at which true future actions yield lower error than actions from another episode (Figure 5, left). pending

Efficiency. Table 4 compares parameters, compute per forecast step and latency. Transport adds pending parameters. Because SHIFTWM and Direct decode all K steps in parallel, a K -step forecast needs one encoder pass, whereas recursive predictors need K (pending).

6 CONCLUSION

We presented SHIFTWM, a latent world model that forecasts by moving observed foundation features: for every future step, a gated local transport of measured patch features plus a learned correction. Across held-out real robot video, surgical robot video, single-image manipulation forecasting and planning benchmarks, pending. The transport field gives a readable view of the predicted motion, and our analysis of action sensitivity shows what anchoring costs and how to recover it for control.

Table 2: **Ablations on DROID validation** (standardised MSE, avg / final step ↓, and action ranking accuracy ↑). Each row changes one component of the default SHIFTWM.

	Variant	Avg	K	Rank acc.
	SHIFTWM (window 7, 3 sources, scaled correction)	pending	pending	pending
Transport	no transport (= Direct)	pending	pending	pending
	global window	pending	pending	pending
	window $w = 3 / w = 11$	pending	pending	pending
Memory	last frame only ($S=1$)	pending	pending	pending
Correction	tanh-bounded	pending	pending	pending
	none (pure transport)	pending	pending	pending
Actions	action-free	pending	pending	pending
	+ action-contrastive ($\lambda=1$)	pending	pending	pending
Encoder	DINOv2-B/14	pending	pending	pending
	DINOv3-S/16	pending	pending	pending
	DINOv2-S, 4×4 pooled grid	pending	pending	pending

Table 3: **Planning success (%) at the official protocol** (CEM, 300 samples $\times$ 30 iterations, 50 episodes $\times$ 3 planner seeds). All predictors use frozen DINOv2-S patches. The LeWM row is the released checkpoint (end-to-end encoder) for reference.

Predictor	PushT	TwoRoom	Reacher	Plan time (s)
LeWM (released) (Maes et al., 2026b)	88.0	pending	pending	0.50
AR-TF (DINO-WM-style)	pending	pending	pending	pending
AR (rollout-trained)	pending	pending	pending	pending
Direct	pending	pending	pending	pending
SHIFTWM	pending	pending	pending	pending
SHIFTWM + action-contrastive	pending	pending	pending	pending

Limitations. SHIFTWM predicts features, not pixels, so visualisations and pixel metrics rely on a separately trained decoder. Transport can only copy content that has been observed, so newly revealed regions depend on the correction term. The action-contrastive objective uses in-batch negatives, which can be easy to tell apart in diverse datasets. The surgical benchmark uses scene-camera views of dVRK training tasks, not in-vivo endoscopy.

REPRODUCIBILITY STATEMENT

Code, configurations, per-episode evaluation records and trained checkpoints will be released. Appendix A lists every hyperparameter. Dataset splits are deterministic and documented in Appendix B. All datasets are public: DROID (CC-BY-4.0), Open-H-Embodiment (CC-BY-4.0), the RLA-WM/IWS release, and the LeWM planning datasets (MIT).

ETHICS STATEMENT

The surgical data are public recordings of robotic procedures on phantoms and ex-vivo tissue, released under CC-BY-4.0. We make no clinical claims, and the models are not intended for clinical use.

AI USE STATEMENT

In this work, we used generative AI tools to assist with writing and debugging research code, orchestrating experiments, surveying related work, and editing the manuscript text. All AI-assisted code was tested, and every reported number is produced by logged runs that the authors reviewed. Bibliographic entries were verified against the original sources. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

Pending results. Qualitative grid (DROID and surgical test episodes): observed frame, future ground truth, learned transport field (arrows = expected source displacement per patch), gate map g_k , optical-flow reference (RAFT), and per-patch error maps of AR vs. SHIFWWM, decoded to RGB for steps $k \in \{1, 5, 10\}$.

Figure 4: **The transport field is a readable forecast of motion.** For each held-out example (rows), the arrows show where each predicted patch draws its features from, and colour shows the gate. Transport concentrates on the moving arm, gripper and instruments. The static background keeps the identity path, and newly revealed regions fall back to the correction term. The rightmost columns compare decoded forecasts and error maps with the strongest baseline.

Pending results. Forecast error (x) vs. action ranking accuracy (y) for every predictor and ablation; marker size = planning success.

Pending results. Relative MSE gain of SHIFWWM over Direct, binned by true per-patch feature change, DROID and Hamlyn test.

Figure 5: **Left: accuracy vs. action sensitivity.** Anchored predictors can reach low open-loop error while their forecasts react weakly to the actions. The action-contrastive objective moves SHIFWWM toward the upper-left. **Right: gains concentrate where the scene moves.**

REFERENCES

- Eloi Alonso, Adam Jelley, Vincent Micheli, Anssi Kanervisto, Amos Storkey, Tim Pearce, and François Fleuret. Diffusion for world modeling: Visual details matter in atari. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. URL <https://arxiv.org/abs/2405.12399>.
- Mido Assran, Adrien Bardes, David Fan, Quentin Garrido, Russell Howes, Mojtaba Komeili, Matthew Muckley, Ammar Rizvi, Claire Roberts, et al. V-JEPA 2: Self-Supervised video models enable understanding, prediction and planning. *arXiv preprint arXiv:2506.09985*, 2025. URL <https://arxiv.org/abs/2506.09985>.
- Federico Baldassarre, Marc Szafraniec, Basile Terver, Vasil Khalidov, Francisco Massa, Yann LeCun, Patrick Labatut, Maximilian Seitzer, and Piotr Bojanowski. Back to the features: DINO as a foundation for video world models. *arXiv preprint arXiv:2507.19468*, 2025. URL <https://arxiv.org/abs/2507.19468>.
- Randall Balestriero and Yann LeCun. LeJEPA: Provable and scalable Self-Supervised learning without the heuristics. *arXiv preprint arXiv:2511.08544*, 2025. URL <https://arxiv.org/abs/2511.08544>.
- Amir Bar, Gaoyue Zhou, Danny Tran, Trevor Darrell, and Yann LeCun. Navigation world models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. URL <https://arxiv.org/abs/2412.03572>.

Table 4: **Efficiency** for a 10-step forecast of a 16×16 grid on one H200 (batch 1, bf16). Training cost is for one DROID run.

Predictor	Params (M)	GFLOPs / 10 steps	Latency (ms)	Train GPU-h
AR-TF / AR	21.70	pending	pending	pending
Direct	21.70	pending	pending	pending
SHIFTWM	21.75	pending	pending	pending

Gabrijel Boduljak, Yushi Lan, Christian Rupprecht, and Andrea Vedaldi. VFMMF: World modeling by forecasting vision foundation model features. *arXiv preprint arXiv:2512.11225*, 2025. URL <https://arxiv.org/abs/2512.11225>.

Jake Bruce, Michael Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, 2024. URL <https://arxiv.org/abs/2402.15391>.

Bert De Brabandere, Xu Jia, Tinne Tuytelaars, and Luc Van Gool. Dynamic filter networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016. URL <https://arxiv.org/abs/1605.09673>.

Haoyi Duan, Hong-Xing Yu, Sirui Chen, Li Fei-Fei, and Jiajun Wu. WorldScore: A unified evaluation benchmark for world generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. URL <https://arxiv.org/abs/2504.00983>.

Frederik Ebert, Chelsea Finn, Alex X. Lee, and Sergey Levine. Self-supervised visual planning with temporal skip connections. In *Conference on Robot Learning (CoRL)*, volume 78 of *PMLR*, 2017. URL <https://arxiv.org/abs/1710.05268>.

Chelsea Finn, Ian Goodfellow, and Sergey Levine. Unsupervised learning for physical interaction through video prediction. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL <https://arxiv.org/abs/1605.07157>.

Yixin Gao, S. Swaroop Vedula, Carol E. Reiley, Narges Ahmadi, Balakrishnan Varadarajan, Henry C. Lin, Lingling Tao, Luca Zappella, Benjamín Béjar, David D. Yuh, Chi Chiung Grace Chen, René Vidal, Sanjeev Khudanpur, and Gregory D. Hager. JHU-ISI gesture and skill assessment working set (JIGSAWS): A surgical activity dataset for human motion modeling. In *Modeling and Monitoring of Computer Assisted Interventions (M2CAI), MICCAI Workshop*, 2014. URL https://cirl.lcsr.jhu.edu/research/hmm/datasets/jigsaws_release/.

Jun Guo, Xiaojian Ma, Yikai Wang, Min Yang, Huaping Liu, and Qing Li. FlowDreamer: A RGB-D world model with flow-based motion representations for robot manipulation. *IEEE Robotics and Automation Letters*, 2026a. doi: 10.1109/LRA.2026.3653273. URL <https://arxiv.org/abs/2505.10075>.

Yanjiang Guo, Lucy Xiaoyang Shi, Jianyu Chen, and Chelsea Finn. Ctrl-World: A controllable generative world model for robot manipulation. In *International Conference on Learning Representations (ICLR)*, 2026b. URL <https://arxiv.org/abs/2510.10125>.

Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *PMLR*, 2019. URL <https://arxiv.org/abs/1811.04551>.

Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. doi: 10.1038/s41586-025-08744-2. URL <https://arxiv.org/abs/2301.04104>.

Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.16828>.

486 Jesse Haworth, Juo-Tung Chen, Nigel Nelson, Ji Woong Kim, Masoud Moghani, Chelsea Finn,
487 and Axel Krieger. SutureBot: A precision framework & benchmark for autonomous End-to-
488 End suturing. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and*
489 *Benchmarks Track*, 2025. URL <https://arxiv.org/abs/2510.20965>.

490

491 Yufan He, Pengfei Guo, Mengya Xu, Zhaoshuo Li, Andriy Myronenko, Dillan Imans, Bingjie Liu,
492 Dongren Yang, Mingxue Gu, Yongnan Ji, et al. Cosmos-H-Surgical: Learning surgical robot
493 policies from videos via world modeling. *arXiv preprint arXiv:2512.23162*, 2025. URL <https://arxiv.org/abs/2512.23162>.

494

495 Siqiao Huang, Partha Kaushik, Michael Chen, Hengkai Pan, Kaiwen Geng, Omar Chehab, Fernando
496 Moreno-Pino, and Max Simchowitz. Nano world models: A minimalist implementation of future
497 video prediction. *arXiv preprint arXiv:2605.23993*, 2026. URL <https://arxiv.org/abs/2605.23993>.

498

499

500 Efstathios Karypidis, Ioannis Kakogeorgiou, Spyros Gidaris, and Nikos Komodakis. DINO-
501 Foresight: Looking into the future with DINO. In *Advances in Neural Information Processing*
502 *Systems (NeurIPS)*, 2025. URL <https://arxiv.org/abs/2412.11673>.

503

504 Alexander Khazatsky, Karl Pertsch, Suraj Nair, Ashwin Balakrishna, Sudeep Dasari, Siddharth
505 Karamcheti, Soroush Nasiriany, Mohan Kumar Srirama, Lawrence Yunliang Chen, Kirsty El-
506 lis, et al. DROID: A large-scale in-the-wild robot manipulation dataset. In *Robotics: Science and*
507 *Systems (RSS)*, 2024. URL <https://arxiv.org/abs/2403.12945>.

508

509 Moo Jin Kim, Yihuai Gao, Tsung-Yi Lin, Yen-Chen Lin, Yunhao Ge, Grace Lam, Percy Liang,
510 Shuran Song, Ming-Yu Liu, Chelsea Finn, and Jinwei Gu. Cosmos policy: Fine-Tuning video
511 models for visuomotor control and planning. *arXiv preprint arXiv:2601.16163*, 2026. URL
<https://arxiv.org/abs/2601.16163>.

512

513 Saurabh Koju, Saurav Bastola, Prashant Shrestha, Sanskar Amgain, Yash Raj Shrestha, Rudra P. K.
514 Poudel, and Binod Bhattarai. Surgical vision world model. In *Data Engineering in Medical*
515 *Imaging (DEMI), MICCAI Workshops*, Lecture Notes in Computer Science, 2025. doi: 10.1007/
516 978-3-032-08009-7_1. URL <https://arxiv.org/abs/2503.02904>.

517

518 Alex X. Lee, Richard Zhang, Frederik Ebert, Pieter Abbeel, Chelsea Finn, and Sergey Levine.
519 Stochastic adversarial video prediction. *arXiv preprint arXiv:1804.01523*, 2018. URL <https://arxiv.org/abs/1804.01523>.

520

521 Chenxin Li, Hengyu Liu, Yifan Liu, Brandon Y. Feng, Wuyang Li, Xinyu Liu, Zhen Chen, Jing
522 Shao, and Yixuan Yuan. Endora: Video generation models as endoscopy simulators. In *Medical*
523 *Image Computing and Computer Assisted Intervention (MICCAI)*, volume 15006 of *Lecture Notes*
524 *in Computer Science*, pp. 230–240, 2024. doi: 10.1007/978-3-031-72089-5_22. URL <https://arxiv.org/abs/2403.11050>.

525

526 Dacheng Li, Yunhao Fang, Yukang Chen, Shuo Yang, Shiyi Cao, Justin Wong, Michael Luo, Xiao-
527 long Wang, Hongxu Yin, Joseph E. Gonzalez, et al. WorldModelBench: Judging video genera-
528 tion models as world models. In *Advances in Neural Information Processing Systems (NeurIPS),*
529 *Datasets and Benchmarks Track*, 2025. URL <https://arxiv.org/abs/2502.20694>.

530

531 Huanrong Liu, Weiliang Huang, Bob Zhang, Weichao Cai, Chunlin Tian, and Qingbiao Li. Surg-
532 WMBench: A Vision-Based benchmark for World-Modeling surgical instrument motion planning.
533 *arXiv preprint arXiv:2608.08070*, 2026. URL <https://arxiv.org/abs/2608.08070>.

534

535 Ziwei Liu, Raymond A. Yeh, Xiaoou Tang, Yiming Liu, and Aseem Agarwala. Video frame syn-
536 thesis using deep voxel flow. In *Proceedings of the IEEE International Conference on Computer*
537 *Vision (ICCV)*, 2017. URL <https://arxiv.org/abs/1702.02463>.

538

539 Po-Chien Luan, Zimin Xia, Wuyang Li, Yang Gao, and Alexandre Alahi. Drift-Resistant navigation
world model with anchored epipolar guidance. *arXiv preprint arXiv:2605.24761*, 2026. URL
<https://arxiv.org/abs/2605.24761>.

- Lucas Maes, Quentin Le Lidec, Luiz Facury, Nassim Massaudi, Ayush Chaurasia, Francesco Capuano, Richard Gao, Taj Gillin, Dan Haramati, Damien Scieur, Yann LeCun, and Randall Balestriero. stable-worldmodel: A platform for reproducible world modeling research and evaluation. *arXiv preprint arXiv:2605.21800*, 2026a. URL <https://arxiv.org/abs/2605.21800>.
- Lucas Maes, Quentin Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. Leworld-model: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026b. URL <https://arxiv.org/abs/2603.19312>.
- Vincent Micheli, Eloi Alonso, and François Fleuret. Transformers are Sample-Efficient world models. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2209.00588>.
- Malte Mosbach, Jan Niklas Ewertz, Angel Villar-Corrales, and Sven Behnke. SOLD: Slot Object-Centric latent dynamics models for relational manipulation learning from pixels. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, 2025. URL <https://arxiv.org/abs/2410.08822>.
- Saman Motamed, Laura Culp, Kevin Swersky, Priyank Jaini, and Robert Geirhos. Do generative video models understand physical principles? In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026. doi: 10.1109/WACV61042.2026.00099. URL <https://arxiv.org/abs/2501.09038>.
- Nilaksh, Saurav Jha, Artem Zhohus, and Sarath Chandar. Reconstruction or semantics? what makes a latent space useful for robotic world models. *arXiv preprint arXiv:2605.06388*, 2026. URL <https://arxiv.org/abs/2605.06388>.
- NVIDIA. Cosmos world foundation model platform for physical AI. *arXiv preprint arXiv:2501.03575*, 2025a. URL <https://arxiv.org/abs/2501.03575>.
- NVIDIA. World simulation with video foundation models for physical AI. *arXiv preprint arXiv:2511.00062*, 2025b. URL <https://arxiv.org/abs/2511.00062>.
- Chinedu Innocent Nwoye, Tong Yu, Cristians Gonzalez, Barbara Seeliger, Pietro Mascagni, Didier Mutter, Jacques Marescaux, and Nicolas Padoy. Rendezvous: Attention mechanisms for the recognition of surgical action triplets in endoscopic videos. *Medical Image Analysis*, 78:102433, 2022. URL <https://arxiv.org/abs/2109.03223>.
- Open-H-Embodiment Consortium. Open-H-Embodiment: A Large-Scale dataset for enabling foundation models in medical robotics. *arXiv preprint arXiv:2604.21017*, 2026. URL <https://arxiv.org/abs/2604.21017>.
- Open X-Embodiment Collaboration. Open X-Embodiment: Robotic learning datasets and RT-X models. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024. doi: 10.1109/ICRA57147.2024.10611477. URL <https://arxiv.org/abs/2310.08864>.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL <https://arxiv.org/abs/2304.07193>.
- Wentao Pan, Wuyang Li, Shengyuan Liu, Xinyu Liu, Hengyu Liu, and Yixuan Yuan. SurgVista: Long-Horizon surgical world modeling with plausible Instrument-Tissue dynamics. *arXiv preprint arXiv:2606.19889*, 2026. URL <https://arxiv.org/abs/2606.19889>.
- Jack Parker-Holder and Shlomi Fruchter. Genie 3: A new frontier for world models. Google DeepMind blog, August 2025. URL <https://deepmind.google/discover/blog/genie-3-a-new-frontier-for-world-models/>.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4195–4205, 2023.

- Francois Porcher, Nicolas Carion, Karteek Alahari, and Shizhe Chen. Flow matching in feature space for stochastic world modeling. *arXiv preprint arXiv:2606.29059*, 2026. URL <https://arxiv.org/abs/2606.29059>.
- Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, Cijo Jose, Vasil Khalidov, Marc Szafraniec, Seungeun Yi, Michaël Ramamonjisoa, et al. DINOv3. *arXiv preprint arXiv:2508.10104*, 2025. URL <https://arxiv.org/abs/2508.10104>.
- Ssharvien Kumar Sivakumar, Akwele Johnson, Anirudh Dhingra, Yannik Frisch, Ghazal Ghazaei, and Anirban Mukhopadhyay. SWoMo: Neuro-symbolic world model for cataract surgery simulation. *arXiv preprint arXiv:2605.16530*, 2026. URL <https://arxiv.org/abs/2605.16530>.
- Vlad Sobal, Wancong Zhang, Kyunghyun Cho, Randall Balestriero, Tim G. J. Rudner, and Yann LeCun. Learning from Reward-Free offline data: A case for planning with latent dynamics models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL <https://arxiv.org/abs/2502.14819>.
- Shahram Najam Syed, Arthur Jakobsson, Haoran Hao, and Jeffrey Ichnowski. Intercepting the future: Latent-Space predictive world model for dynamic VLA manipulation. *arXiv preprint arXiv:2606.02486*, 2026. URL <https://arxiv.org/abs/2606.02486>.
- Zachary Teed and Jia Deng. RAFT: Recurrent All-Pairs field transforms for optical flow. In *European Conference on Computer Vision (ECCV)*, 2020. doi: 10.1007/978-3-030-58536-5_24. URL <https://arxiv.org/abs/2003.12039>.
- Basile Terver, Tsung-Yen Yang, Jean Ponce, Adrien Bardes, and Yann LeCun. What drives success in physical planning with Joint-Embedding predictive world models? *Transactions on Machine Learning Research*, 2026. URL <https://arxiv.org/abs/2512.24497>.
- Andru P. Twinanda, Sherif Shehata, Didier Mutter, Jacques Marescaux, Michel de Mathelin, and Nicolas Padoy. EndoNet: A deep architecture for recognition tasks on laparoscopic videos. *IEEE Transactions on Medical Imaging*, 36(1):86–97, 2017. doi: 10.1109/TMI.2016.2593957. URL <https://arxiv.org/abs/1602.03012>.
- Vikram Voleti, Alexia Jolicoeur-Martineau, and Christopher Pal. MCVD: Masked conditional video diffusion for prediction, generation, and interpolation. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL <https://arxiv.org/abs/2205.09853>.
- Carl Vondrick and Antonio Torralba. Generating the future with adversarial transformers. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017. URL https://openaccess.thecvf.com/content_cvpr_2017/html/Vondrick_Generating_the_Future_CVPR_2017_paper.html.
- Homer Walke, Kevin Black, Abraham Lee, Moo Jin Kim, Max Du, Chongyi Zheng, Tony Zhao, Philippe Hansen-Estruch, Quan Vuong, Andre He, et al. BridgeData V2: A dataset for robot learning at scale. In *Conference on Robot Learning (CoRL)*, 2023. URL <https://arxiv.org/abs/2308.12952>.
- Yixuan Wang, Rhythm Syed, Fangyu Wu, Mengchao Zhang, Aykut Onol, Jose Barreiros, Hooshang Nayyeri, Tony Dear, Huan Zhang, and Yunzhu Li. Interactive world simulator for robot policy training and evaluation. *arXiv preprint arXiv:2603.08546*, 2026. URL <https://arxiv.org/abs/2603.08546>.
- Ziyi Wu, Nikita Dvornik, Klaus Greff, Thomas Kipf, and Animesh Garg. SlotFormer: Unsupervised visual dynamics simulation with Object-Centric models. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2210.05861>.
- Sherry Yang, Yilun Du, Kamyar Ghasemipour, Jonathan Tompson, Leslie Kaelbling, Dale Schuurmans, and Pieter Abbeel. Learning interactive Real-World simulators. In *International Conference on Learning Representations (ICLR)*, 2024. URL <https://arxiv.org/abs/2310.06114>.

-
- Jewon Yeom, Hanseul Kim, Jeongjae Park, Sungmok Jung, Jaejin Lee, and Taesup Kim. What makes video world model latents Action-Relevant: Prediction over reconstruction. *arXiv preprint arXiv:2606.07687*, 2026. URL <https://arxiv.org/abs/2606.07687>.
- Shicheng Yin, Kaixuan Yin, Weixing Chen, Yang Liu, Guanbin Li, and Liang Lin. DDP-WM: Disentangled dynamics prediction for efficient world models. *arXiv preprint arXiv:2602.01780*, 2026. URL <https://arxiv.org/abs/2602.01780>.
- Hu Yue, Siyuan Huang, Yue Liao, Shengcong Chen, Pengfei Zhou, Liliang Chen, Maoqing Yao, and Guanghui Ren. EWM Bench: Evaluating scene, motion, and semantic quality in embodied world models. *arXiv preprint arXiv:2505.09694*, 2025. URL <https://arxiv.org/abs/2505.09694>.
- Xinyu Zhang, Zhengtong Xu, Yutian Tao, Yeping Wang, Yu She, and Abdeslam Boularias. Learning visual feature-based world models via residual latent action. *arXiv preprint arXiv:2605.07079*, 2026a. URL <https://arxiv.org/abs/2605.07079>.
- Zhenghao Zhang, Yuanxiang Wang, Zhenyu Guan, Yujia Yang, Bingkang Shi, Tianyu Zong, Hongzhu Yi, Guoqing Chao, Xingchen Chen, Tiankun Yang, et al. Delta-JEPA: Learning Action-Sensitive world models via latent difference decoding. *arXiv preprint arXiv:2606.31232*, 2026b. URL <https://arxiv.org/abs/2606.31232>.
- Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World models on pre-trained visual features enable zero-shot planning. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pp. 79115–79135, 2025. URL <https://arxiv.org/abs/2411.04983>.
- Fangqi Zhu, Hongtao Wu, Song Guo, Yuxiao Liu, Chilam Cheang, and Tao Kong. IRASim: A Fine-Grained world model for robot manipulation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. URL <https://arxiv.org/abs/2406.14540>.

Appendix	Contents
A Implementation	architecture sizes, hyperparameters, training and compute
B Datasets	DROID, Open-H Hamlyn, IWS, planning suites; splits and action spaces
C Baselines	matched predictors, DINO-WM-style, V-JEPA 2-AC reference
D Full results	per-horizon, per-task, per-camera, cosine and pooled metrics
E Ablations	transport window, sources, correction, encoder, horizon, contrastive weight
F Decoded metrics	feature-to-RGB decoder, PSNR/SSIM/LPIPS
G Probing	robot-state and kinematics probes on predicted features
H Planning	protocol, reproduction of LeWM, per-environment results
I Analysis	transport vs. optical flow, gains vs. motion, action sensitivity
J Qualitative	rollouts, transport fields, gates, failure cases
K Statistics	bootstrap procedure and multiple-comparison control

A IMPLEMENTATION DETAILS

Architecture. Width $d=384$, 6 heads, MLP ratio 4, memory encoder $L_e=4$ layers of full self-attention over the $H \cdot N=768$ tokens of $H=3$ observed 16×16 grids. Query decoder: $L_d=4$ layers, each with self-attention over the $N=256$ query tokens, cross-attention to memory and an MLP, modulated by AdaLN-zero on c_k . Action embedding: a 2-layer MLP followed by a single-layer unidirectional GRU of width d . Transport: key dimension 64, window $w=7$ over $S=3$ source frames (147 candidates per query), learned identity bias initialised at $\beta=4$, and gate bias -3 . Correction head: LayerNorm, then a zero-initialised linear map to C , then a per-channel scale initialised at 1. Parameters: 21.75M (SHIFTWM) and 21.70M (Direct, AR, AR-TF).

Training. AdamW ($\beta=(0.9, 0.95)$, weight decay 0.05), peak learning rate 3×10^{-4} with 800 warm-up steps and cosine decay, 16k steps, batch 48, bf16 autocast, gradient clipping at 1.0. Validation MSE (averaged over horizons, windows at stride 2) is computed every 2k steps, and the best checkpoint is used for testing. AR is trained through its full 10-step rollout. AR-TF is trained with teacher forcing and rolled out at test time. All features for a split are kept in GPU memory as float16 and windows are gathered by index.

Table 5: **Hyperparameters** (shared by all learned predictors unless noted).

Width / heads / MLP ratio	384 / 6 / 4	Optimiser	AdamW, $\beta=(0.9, 0.95)$
Memory encoder layers L_e	4	Peak LR / warm-up	3×10^{-4} / 800 steps
Query decoder layers L_d	4	Schedule	cosine, 16k steps
History H / horizon K	3 / 10	Batch size	48 windows
Transport window w / sources S	7 / 3	Weight decay / clip	0.05 / 1.0
Key dim / identity bias init	64 / 4.0	Precision	bf16 autocast
Gate bias init	-3	Selection	best val. MSE (every 2k steps)
Action-contrastive λ / margin	1.0 / 0.05 (planning)	Seeds	0, 1, 2

Compute. All experiments ran on NVIDIA H200 GPUs, packing at most two learned predictors per GPU. pending

B DATASETS AND SPLITS

Table 6: **Datasets.** Step = one model time step. All splits are disjoint at the unit shown.

Dataset	Domain	Episodes (train/val/test)	Step	Action dim	Split unit	License
DROID (24 shards)	real Franka, 3 cams	851 / 143 / 132	0.33 s	35	session	CC-BY-4.0
Open-H Hamlyn (7 tasks)	surgical dVRK	708 / 152 / 153	0.33 s	80	episode	CC-BY-4.0
IWS PushT / Box / Rope	real manipulation	pending	pending	pending	official handles	pending
PushT / TwoRoom / Reacher	simulation	pending	5 env steps	10	episode	MIT

DROID. We use 24 official full-release shards (22.0 GB) containing 1,126 successful episodes. Splits are assigned by recording session with a salted hash, so no session appears in two splits. Frames are subsampled every 5 native steps (15 Hz $\rightarrow$ 3 Hz), and each action block concatenates the 5 commanded Cartesian poses and gripper positions ($5 \times 7=35$).

Open-H-Embodiment Hamlyn. Seven dVRK training tasks, released under CC-BY-4.0 in LeRobot format. We use the `observation.images.color` stream, which the release labels “endoscope” but which is an Intel RealSense scene camera (848×480). One step is 1/3 s: stride 10 at 30 Hz, or stride 5 for the 15 Hz tissue-lifting task. Each action block holds five commands at 15 Hz, each an 8-D pose-and-gripper target for each arm ($5 \times 16=80$). Episodes shorter than 14 steps (6 of 1,019) are removed. Splits are 70/15/15 per task, assigned by a hash of the episode id.

IWS. pending

Planning suites. PushT, TwoRoom and Reacher expert datasets from the LeWM release (HDF5). Episodes are split disjointly into train and validation sets for world-model training. Evaluation follows the official protocol (Appendix H).

C BASELINES

Matched predictors. All learned baselines share SHIFTWM’s memory encoder, decoder, width, depth, optimiser and schedule, and differ only in the output head and rollout. This isolates the contribution of transport from those of capacity and spatial communication. Direct has the same cross-patch attention as SHIFTWM and differs only in lacking the transport and gate.

DINO-WM-style (AR-TF). Following Zhou et al. (2025), the predictor regresses absolute next-step patch features with teacher forcing and is rolled out autoregressively at test time. We use the same frozen DINOv2 patches at native resolution.

External reference: V-JEPA 2-AC. pending

Earlier variant. Our earlier compact variant (4×4 pooled grids, per-patch temporal predictor, tanh-bounded correction) is reported for continuity in Appendix E.

D FULL FORECASTING RESULTS

Table 7: **Per-horizon test MSE** (DROID cam 1, DINOv2-S; mean $\pm$ std over 3 seeds).

Method	$k=1$	2	3	4	5	6	7	8	9	10
Persistence	0.108	0.165	0.205	0.235	0.258	0.278	0.294	0.307	0.317	0.326
AR-TF (DINO-WM-style)	pending	pending	pending	pending	pending	pending	pending	pending	pending	pending
AR (rollout-trained)	pending	pending	pending	pending	pending	pending	pending	pending	pending	pending
Direct (cross-attn)	0.095	0.136	0.164	0.185	0.200	0.214	0.226	0.235	0.241	0.248
SHIFTWM (ours)	0.087	0.125	0.150	0.169	0.183	0.196	0.207	0.216	0.223	0.229

Table 8: **Open-H Hamlyn per-task test MSE** (avg over horizons, DINOv2-S, 3 seeds).

Method	Knot	Needle	Peg	Sut. 1	Sut. 2	Lift	Retract
Persistence	pending	pending	pending	pending	pending	pending	pending
AR-TF (DINO-WM-style)	pending	pending	pending	pending	pending	pending	pending
AR (rollout-trained)	pending	pending	pending	pending	pending	pending	pending
Direct (cross-attn)	pending	pending	pending	pending	pending	pending	pending
SHIFTWM (ours)	pending	pending	pending	pending	pending	pending	pending

pending

Table 9: **IWS per-task results** on the official RLA-WM handles (feature MSE, decoded LPIPS).

Method	PushT		Box		Rope	
	MSE	LPIPS	MSE	LPIPS	MSE	LPIPS
Persistence	pending	pending	pending	pending	pending	pending
AR	pending	pending	pending	pending	pending	pending
Direct	pending	pending	pending	pending	pending	pending
SHIFTWM	pending	pending	pending	pending	pending	pending

E ADDITIONAL ABLATIONS

Pending results. Validation MSE vs. transport window size $w \in \{3, 5, 7, 11, \text{global}\}$ and number of source frames $S \in \{1, 3\}$.

Figure 6: **Transport window and memory size.**

pending

F DECODED-PIXEL METRICS

Table 10: **Decoded-pixel metrics** at $k=10$ (test splits). All methods are decoded by the same feature-to-RGB decoder (one per dataset, trained on true train features).

Method	DROID			Open-H Hamlyn		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Decoder on true features (upper bound)	pending	pending	pending	pending	pending	pending
Persistence	pending	pending	pending	pending	pending	pending
AR-TF (DINO-WM-style)	pending	pending	pending	pending	pending	pending
AR	pending	pending	pending	pending	pending	pending
Direct	pending	pending	pending	pending	pending	pending
SHIFTWM	pending	pending	pending	pending	pending	pending

pending

G PROBING PREDICTED FEATURES

pending

H PLANNING PROTOCOL AND RESULTS

Faithfulness check. With the released LeWM PushT checkpoint and the official protocol, our harness reaches 45/50, 44/50 and 43/50 successes for planner seeds 42/43/44 (mean 88%). An adapter check confirms that our pluggable predictor interface produces exactly the native LeWM planning costs (maximum absolute difference 0). The LeWM paper reports 96 ± 2.8 , averaged over training seeds, whereas the released checkpoint is a single training seed. pending

Table 11: **Linear probes** from predicted features at $k=10$ to robot state (DROID: commanded end-effector position, cm; Hamlyn: measured tip position of both instruments, mm). Ridge probes on mean- and 4×4 -pooled features are fit on true train features; MAE is per coordinate.

Method	DROID EE position		Hamlyn tip position	
	MAE↓	Pearson r ↑	MAE↓	Pearson r ↑
True future features (upper bound)	pending	pending	pending	pending
Persistence	pending	pending	pending	pending
AR-TF (DINO-WM-style)	pending	pending	pending	pending
AR	pending	pending	pending	pending
Direct	pending	pending	pending	pending
SHIFTWM	pending	pending	pending	pending

Table 12: **Official planning protocol** (le-wm@8edfeb33, stable-worldmodel@4821c8e6).

Planner	CEM, 300 samples $\times$ 30 iterations, 30 elites, initial variance 1, warm start
Horizon / replanning	5 action blocks of 5 env steps; replan every 25 steps
Episodes / seeds	50 episodes per planner seed; seeds 42, 43, 44
Goal	dataset frame 25 steps later in the same trajectory
Budget	50 environment steps
Cost	squared distance between final predicted latent and goal latent
Success	PushT: position < 20 px and angle $< \pi/9$; TwoRoom: < 16 px; Reacher: joint match

I ANALYSIS DETAILS

Pending results. Patch-level agreement between the transport field and RAFT optical flow (cosine of displacement, end-point error) vs. horizon; distribution over test episodes.

Figure 7: **Transport vs. optical flow.**

pending

J QUALITATIVE RESULTS

K STATISTICAL PROCEDURES

Each test episode contributes the mean of its windows (stride 2). For each method and seed we average over episodes, and tables report the mean over three training seeds. For a paired comparison between methods A and B, we compute per-episode differences (averaged over seeds) and draw 10,000 bootstrap resamples of episodes to form a percentile 95% interval. For DROID, we resample sessions instead of episodes to respect the session structure. Primary contrasts (SHIFTWM against each baseline on each dataset) are Holm-adjusted within each dataset.

Pending results. Extended DROID gallery: 6 random test episodes; observed frame, ground truth and decoded forecasts from AR, Direct and SHIFTWM at $k \in \{1, 3, 5, 10\}$, transport arrows and gates.

Figure 8: **DROID test rollouts (random selection).**

Pending results. Extended Open-H Hamlyn gallery: one random test episode per task (7 tasks).

Figure 9: **Surgical test rollouts (one per task, random selection).**

Pending results. Largest-error test windows for SHIFTWM: disocclusion, fast motion, contact events; with transport fields and gates.

Figure 10: **Failure cases.**